

Question 1

- (a) With the goal of summing all integer values in an n cell array, demonstrate the logic required using both “pseudocode” and “flowchart” approaches and demonstrating how each of these representations differ? Make specific reference to the advantages and disadvantages of using “pseudocode” or “flowchart” techniques.

(10 marks)

- (b) Euclid was an ancient Greek mathematician, often referred to as the “Father of Geometry”. In his work *Elements* (c. 300 BC), he proposed an efficient algorithm to compute the greatest common divisor of two numbers (i.e. the largest number that divides both of them without leaving a remainder). An outline of Euclid’s algorithm is provided below:

$$\text{gcd}(a, b) = \begin{cases} a & \text{if } b = 0 \\ \text{gcd}(b, a \bmod b) & \text{otherwise} \end{cases}$$

Where $\text{gcd}(a, b)$ is a method that returns the greatest common divisor, and $a \bmod b$ returns the remainder of the division of a by b .

Outline an algorithm that takes two integers and computes their greatest common divisor using a **recursive implementation** of Euclid’s algorithm, and outputs the answer.

With the aid of this example, show in detail how a **non-recursive** implementation would differ from the recursive approach.

(10 marks)

- (c) Operator precedence plays an important role in how we represent equations in our code.

With the aid of the following short examples, show how operator precedence is correctly applied in each case to determine the correct sequence of operations and the overall correct answer. (Each step of the process must be clearly shown)

- $3 * (2 * 2 + 3) - (4 - (5 ^ 2) * 3)$
- $(18 + ((8) / 4) - (4))$

(5 marks)

PTO

Question 2

- (a) Merge sort is a well known sorting algorithm. With the aid of a detailed example and pseudocode you are asked to outline its features, logical implementation and Big O complexity.

(12 marks)

- (b) Linear Search and Binary Search are two well known searching algorithms. With the aid of a detailed example using the values shown below, and pseudocode, you are asked to outline their features, logical implementation and complexity of both Linear and Binary Search Algorithms.

3	6	9	8	10	15	32	86	98	100
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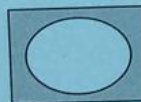
(13 marks)

Question 3

- (a) Explain with the aid of a code or pseudocode example, what is meant by the use of recursion? Using this example, can you demonstrate how recursion is resolved through using the stack?

(5 Marks)

- (b) With the aid of *pseudocode* and a top down *HIPO chart* demonstrate how you would solve the following problem where you are asked to calculate the area remaining when a circle is placed inside a square. You are given the size of the square s and the radius of the sphere r . The area required is shown in grey.
Hint: The area of a square is the (width $\times$ height), while the area of a circle is $(3.14 \times \text{Radius} \times \text{Radius})$



(7 Marks)

- (c) A palindrome is a sequence of characters or numbers that looks the same forwards and backwards. For example, "A Santa at Nasa." is a palindrome because it is spelled the same reading it from front to back as from back to front. The number 1234321 is a numerical palindrome. Write a function that takes a number and its length as arguments and recursively determines whether the number is a palindrome.

(13 marks)

PTO

Question 4:

- (a) Quick sort is a well known sorting algorithm. With the aid of a detailed example and pseudocode you are asked to outline its features, logical implementation and Big O complexity. (13 marks)
- (b) With the aid of a suitable example, demonstrate the benefits of using trees, and what makes them more efficient for certain tasks? (12 marks)

Question 5:

- (a) With the aid of a suitable example, can you demonstrate how a less efficient sorting algorithm differs from a more efficient algorithm in terms of its logical structure and resulting Big O Complexity? (15 marks)
- (b) With the aid of brief examples, outline what is meant by each of the following terms:
- HIPO Charts
 - Modularity
 - Pseudocode
 - Binary Search
 - BOMDAS
- (10 marks)

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